

AI-POWERED MULTILINGUAL COMMUNICATION FOR PHYSIOTHERAPY CONSULTATIONS

CHALLENGE OWNER

The National University Hospital (NUH) is Singapore's leading university hospital and a key member of the National University Health System (NUHS), one of Singapore's three public healthcare clusters. This challenge is jointly led by NUH's Physiotherapy discipline and the Kent Ridge Office of Innovation (KROI).

The Physiotherapy discipline delivers care across specialist outpatient clinics, inpatient wards, and telehealth services. Physiotherapists conduct assessments, consultations, and treatment sessions that rely heavily on effective verbal communication with patients from diverse linguistic backgrounds on a daily basis.

KROI drives innovation across NUH by empowering healthcare professionals to embrace new ways of working, leverage technology, and deliver better patient outcomes. Through its three focus areas — building innovation capabilities, fostering an active innovation ecosystem, and scaling impactful technologies — KROI supports the development and implementation of solutions to address real-world healthcare challenges.

CONTEXT

Physiotherapists rely heavily on verbal interaction to conduct assessments, explain clinical concepts, provide treatment instructions, and educate patients on their conditions and rehabilitation plans. These interactions take place across specialist outpatient clinics, inpatient wards, and increasingly through telehealth consultations.

Language barriers arise in an estimated 30–40% of patient interactions, posing significant challenges in a discipline where nuanced communication, active patient participation, and clear understanding of instructions are essential for safe and effective treatment. A typical physiotherapy consultation involves a range of communication-intensive activities, including:

- Physical assessments requiring patients to perform movements such as squatting, balancing, standing, or sitting while clinicians observe their natural responses.
- Verbal assessments involving detailed questioning about symptoms, daily activities, functional limitations, or mobility needs.
- Safety assessments, particularly for telehealth consultations, where clinicians must verify patient location to ensure appropriate emergency response measures can be activated if required.
- Patient education, including explanations of diagnoses, treatment plans, exercise programmes, and expected recovery outcomes.

The following key challenges have been identified in the current workflow:

1. Language Barriers Affecting Patient Understanding, Engagement, and Adherence

Patients with limited English proficiency often rely on caregivers or translators during consultations, reducing opportunities for physiotherapists to engage directly and ask detailed follow-up questions. This affects the accuracy of assessments, the quality of documentation, and patients' understanding of their condition and treatment plan. Limited understanding reduces patient buy-in and adherence to prescribed exercises, ultimately impacting treatment effectiveness.

2. Difficulty Communicating Clinical Nuances and Healthcare-Specific Concepts

Physiotherapy consultations frequently involve clinical concepts, treatment approaches, and safety considerations that are difficult to translate accurately. Examples include explaining differences between clinical concepts such as aerobic and anaerobic training, discussing symptoms and functional limitations, assessing suitability for mobility aids, and providing exercise instructions where small differences in wording can significantly affect patient performance and safety.

Current translation tools such as Google Translate and MediBabble are largely designed for general communication and lack healthcare-specific contextual understanding. Translation inaccuracies remain common — for example, "cardiac arrest" has been translated as "imprisonment of heart" in Marathi — increasing the risk of misinterpretation, incorrect exercise execution, patient safety risks, or unnecessary anxiety arising from misunderstandings of clinical information.

3. Communication Workarounds that Disrupts Clinical Workflow

Consultations are delayed while clinicians search for available translators who are usually their fellow colleagues, switch between translation tools such as Google Translate and MediBabble, or manage conversations involving multiple parties. Clinicians may also need to repeat or rephrase questions multiple times, with limited visibility into whether information is being communicated accurately. These workarounds increase consultation duration and reduce patient throughput.

4. Fragmented and Inconsistent Communication Approaches

There is currently no standardised communication solution across physiotherapy services. Different clinicians rely on varying combinations of official translators, informal support from multilingual colleagues and caregivers, generic translation applications, and pre-prepared translated materials. Existing resources primarily support one-way communication from clinician to patient, with limited flexibility for follow-up questions or patient responses.

As a result, the quality of care can become dependent on the communication methods available during a particular consultation rather than a consistent standard of delivery. As NUH looks to improve care delivery and expand its telehealth services, effective communication across language barriers is becoming increasingly important. Telehealth has the potential to improve accessibility for patients who prefer rehabilitation in familiar environments, reside overseas, or are advised to minimise hospital visits — but this requires communication approaches that maintain clinical accuracy, patient safety, and patient understanding regardless of language proficiency.

CHALLENGE STATEMENT

How might we enable effective, real-time two-way communication between physiotherapists and patients with limited English proficiency, to improve patient understanding, participation, and adherence to treatment plans, whilst minimising disruption to clinical workflows?

WHAT ARE WE LOOKING FOR?

NUH is seeking a solution that enables seamless, real-time communication between physiotherapists and patients with limited English proficiency. The solution should support accurate translation of healthcare conversations while fitting naturally into existing clinical workflows, minimising disruption to consultations, and supporting the communication of healthcare-specific concepts without replacing the physiotherapist's clinical judgement or decision-making. While the POC will primarily be tested in in-person consultations, the solution should also demonstrate its ability to support telehealth consultations safely and effectively. During telehealth consultations, the solution will continue to be accessed through the clinician's phone, played through the device speaker.

The proposed solution should include the following capabilities:

- Real-Time, Two-Way Clinical Communication. The ability to convert spoken input from both physiotherapists and patients into transcribed and translated text and audio outputs in real time. The solution should accurately translate healthcare and physiotherapy-related terminology while functioning strictly as a communication aid rather than a clinical decision-making tool.
- Continuous Learning and Content Management. The solution should improve translation accuracy over time by learning from clinician interactions. Authorised users should be able to directly refine translations and build a healthcare-specific knowledge base relevant to physiotherapy practice.
- Language Support. For the Proof of Concept, the solution must ideally be able to support Bahasa Indonesia, Bahasa Melayu, Tamil, Hokkien, Cantonese, and Burmese. The solution should also demonstrate the ability to scale to additional languages required for full implementation, including Arabic, Bengali, and other dialects where feasible.
- User Management and Permission Controls. For this proof-of-concept, NUH will provide mock data sets (text & speech) and clinical scenarios to validate the solution's effectiveness. Conversation recording, transcript storage, and playback functionality are not required for the POC, and may be explored in future phases following additional security and governance review.

OVERALL PERFORMANCE REQUIREMENTS

- Hands-Free, Mobile-First Workflow. The solution should operate primarily on physiotherapists' personal mobile devices across all operating systems (iOS & Android), supporting hands-free or near hands-free use to minimise workflow disruption and meet infection control requirements in clinical environments.
- Real-Time Performance. Translation should occur with sufficiently low latency to support natural conversation flow and maintain the efficiency of consultations.
- Security and Data Management. The solution must not be used or retain patient identifiers for learning or model training purposes. Solution providers should clearly describe their data processing architecture, security controls, storage practices, and data retention policies, including whether translation occurs on-device, on-premises, or in the cloud.
- Future Integration Potential (Optional). While integration is not required for the POC, the solution should ideally be designed to integrate with existing healthcare applications and platforms in the future, including telehealth platforms, clinical documentation tools, and government cloud infrastructure. Solutions demonstrating this capability will be viewed favourably.

Challenge solvers who choose to apply for this challenge must be registered and operating in Singapore. The prototype needs to be demonstrated in Singapore.

METRICS OF SUCCESS

The success of the prototype will be evaluated based on its ability to improve both patient care and operational outcomes including:

- Enhanced Patient Engagement. To increase direct patient-clinician interactions by reducing reliance on caregivers or ad-hoc translators during consultations. Success will be reflected in observable improvements in the quality and depth of patient-clinician communication across multilingual consultations.
- Improved Patient Understanding and Treatment Adherence. To improve patients' comprehension of their condition, treatment plan, and prescribed exercises. Success may be indicated by improved comprehension during verbal checks, increased adherence to

prescribed exercises, reduced complaints arising from miscommunication, and improvements in relevant patient-reported outcome measures (PROMs).

- Reduced Consultation Duration. To bring multilanguage consultation durations closer to standard session lengths, which average an hour for the first session and up to 40 minutes for follow-ups. Current sessions requiring language workarounds can take close to twice the duration of a standard session, and the solution should demonstrably reduce this gap.
- Better Manpower Utilisation and Cross-Coverage. To reduce dependence on specific physiotherapists due to language capabilities, enabling a broader pool of clinicians to support patients with diverse language needs. Success will be reflected in improved scheduling flexibility, greater cross-coverage among clinicians, and reduced operational constraints linked to language matching.
- Increased Readiness for Telehealth Delivery. To support effective communication across both in-person and telehealth consultations, enabling broader adoption of remote physiotherapy services. The POC will primarily be tested in in-person consultations, but the solution should also demonstrate its ability to support telehealth consultations safely and effectively, laying the foundation for broader adoption of remote physiotherapy services.

POSSIBLE USE CASES

1. Seamless Multilingual Physiotherapy Consultation. Sarah, a physiotherapist, is seeing a patient who primarily speaks Bahasa Indonesia. After identifying the patient's preferred language, Sarah launches the solution on her mobile device and selects the relevant language pair. The solution automatically captures spoken conversations from both Sarah and the patient, transcribing speech into text, translating it, and delivering translated audio responses in real time. Throughout the consultation, Sarah and the patient communicate naturally. When the patient expresses confusion about an exercise instruction, the solution translates the patient's response and Sarah rephrases her explanation accordingly. Where highly technical medical terminology is used, the solution suggests simpler alternatives for Sarah to consider. The entire interaction runs seamlessly in the background, allowing Sarah to focus on patient engagement rather than managing the technology — all without the need for a separate translator or interpreter.
2. Continuous Improvement of Clinical Translation Knowledge. After her clinic session, Sarah logs into the solution using her authorised account. She notices that a physiotherapy term encountered during the consultation could be translated more accurately and provides feedback through a natural language conversation with the system. She also adds several commonly used phrases and treatment explanations through voice or text input. Over time, the solution learns from clinician-approved inputs and builds a richer physiotherapy-specific knowledge base. This enables future consultations to deliver more accurate translations of clinical concepts, exercise instructions, and patient education content — improving communication quality consistently across the entire physiotherapy team.

WHAT'S IN IT FOR YOU

- SGD50,000 of prize money to develop a Working Prototype, with the scope to be discussed and agreed upon between Challenge Owner and selected Solver, for each winner of this challenge (see Award Model)
- Access to IMDA's PIXEL corporate innovation hub and complimentary innovation consultancies (e.g. Design Thinking, Digital Storytelling) for the prototype development, where applicable
- While the POC will be conducted within NUH's Physiotherapy discipline, the challenge has significant scaling potential across the healthcare ecosystem. Successful solutions may be extended to other allied health professions, clinical specialties, and administrative functions

within NUH, with the opportunity for subsequent deployment across other institutions within the National University Health System (NUHS) cluster.

EVALUATION CRITERIA

The evaluation process shall take place over two stages. Proposals shall be evaluated based on the evaluation criteria set out for the first stage. Thereafter, shortlisted proposals shall be subjected to a second stage evaluation in the form of an interview / pitch, and the scoring shall be based on a re-defined assessment criteria for the selection of the challenge finalist(s).

Solution Fit (30%)	<u>Relevance (30%)</u> : To what extent does the proposed solution address the challenge statement effectively?
Solution Readiness (30%)	<u>Maturity (15%)</u> : How ready is the proposed solution to go to the market? <u>Scalability (15%)</u> : Is there any evidence to suggest capacity to scale?
Solution Advantage (20%)	<u>Quality of Innovation (20%)</u> : Is the solution cost effective and truly innovative? Does it make use of new technologies in the market, and can it potentially generate new IP?
Company Profile (20%)	<u>Business Traction (10%)</u> : Does the company have existing customers and revenue traction? <u>Team Experience (10%)</u> : Do the team members possess strong scientific/technical background?

AWARD MODEL

30% of the prize money will be awarded to each selected finalist at the start of the POC/prototype development process. The remaining 70% will be awarded after completion of the POC/prototype solution, based on milestones agreed between Challenge Owner(s) and the solver. Prize money will be inclusive of any applicable taxes and duties that any of the parties may incur.

*Note that a finalist who is selected to undertake the prototype will be required to enter into an agreement with Challenge Owner(s) that will include more detailed conditions pertaining to the POC/prototype.

SUBMISSION GUIDELINES AND DEADLINE

All submissions should include sufficient information for assessment against the stated evaluation criteria. Challenge Solvers are encouraged to include:

- **Solution proposal** addressing the challenge statement (and any demos, where applicable)
- **Company business profile**, including past projects or relevant track record
- **Team profile** - information on the core team members involved in the project, their roles, technical certifications and relevant experience
- **Prototype development timeline** - outlining key milestones for prototype development and deployment
- **Scalability considerations**, including business model or estimated costs for future deployment, ongoing support and maintenance

All submissions must be made by **14 Aug 2026, 1600 hours (SGT/GMT +8)**. Challenge Owner(s) and IMDA may extend the deadline of the submission at their discretion. Late submissions on the OIP, or submissions via GeBIZ, will not be considered.